L→ 1536

Minimum Swaps to Arrange a Binary Grid :-

①

0	0	1
1	1	0
1	0	0

→ ②

1	0	0
0	0	1
1	1	0

→

1	0	0
1	1	0
0	0	1

→ ② ③

②

1	0	0	0
0	0	0	0
0	1	0	0
0	0	0	0

→

	-	-	-
		✓	✓
			✓

→ ⓪

1	1	1	1
1	1	0	0
0	0	0	0
1	1	1	1

→ ⓪ ⓪ ⓪ ⓪ ✓
1 1 0 0 ✓
1 1 1 1 ✓
1 1 1 1

-1

-1

5×5

③

1	1	0	0
1	1	1	1
0	0	0	0
0	0	1	0

→

0	0	0	0
1	1	0	0
1	1	1	1
0	0	1	0

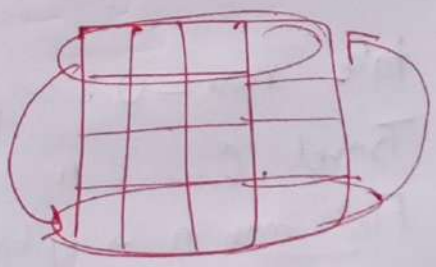
→

0	0	0	0
1	1	0	0
0	0	1	0
1	1	1	1

→ ②

② ①

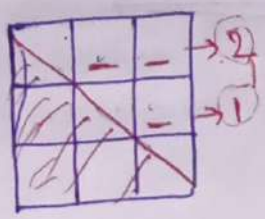
⇒ Important Observations :-



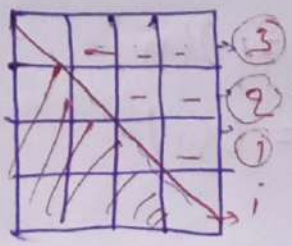
2D ⟶ 1D

•) Grid n×n

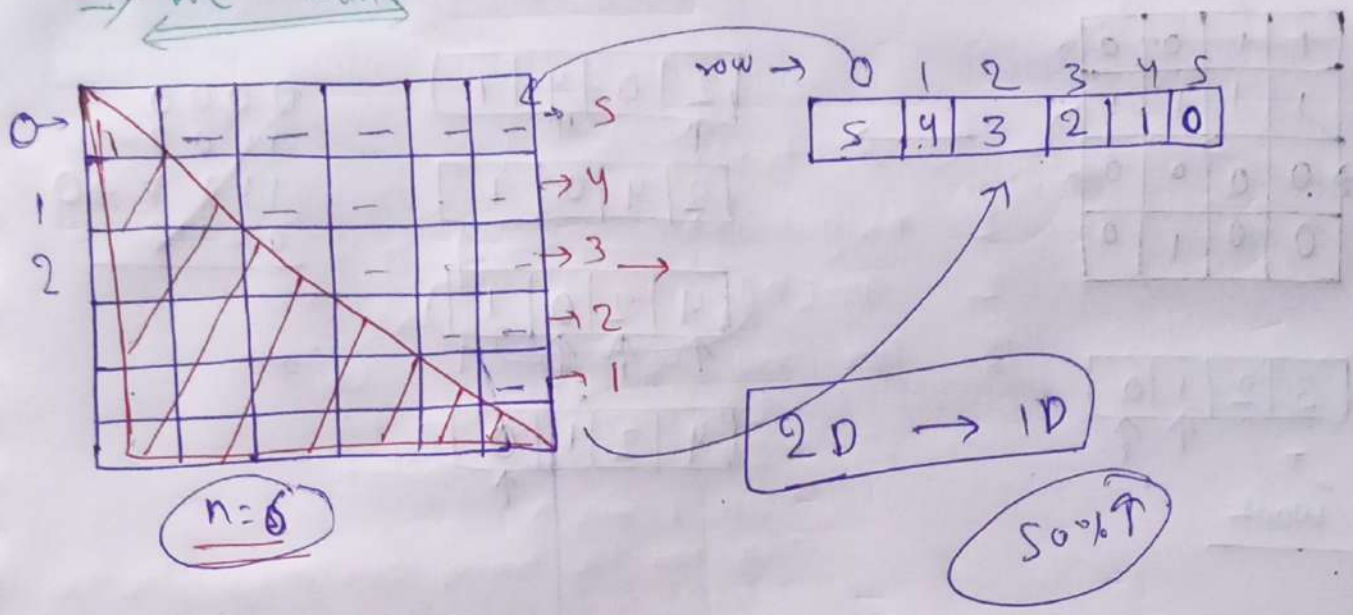
n = 3

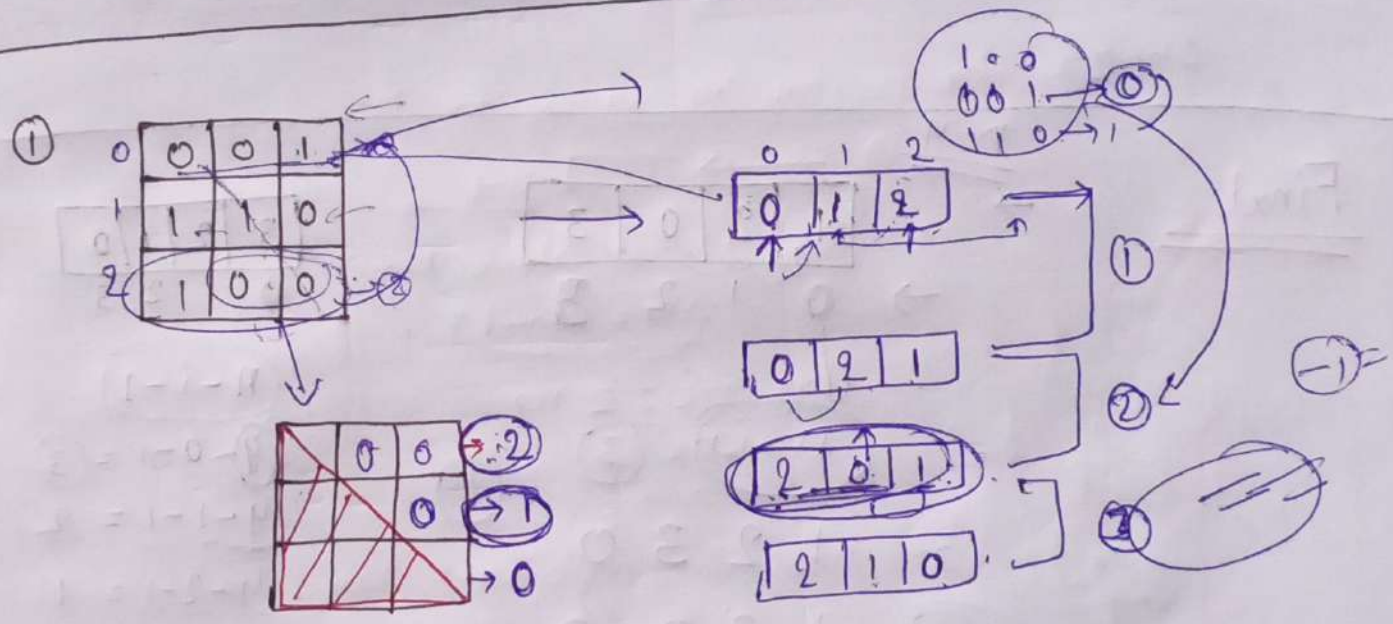


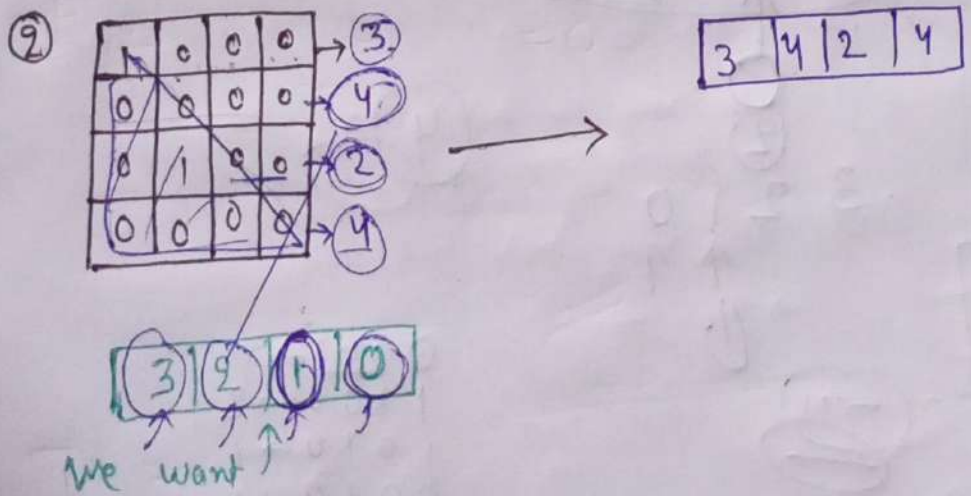
n=4



⇒ We Want


row → 0 1 2 3 4 5
5 4 3 2 1 0
2D → 1D
n=6
50%↑


①
0 1 2
0 2 1
2 0 1
2 1 0
-1


②
3 4 2 4
3 2 1 0
we want
n=4

③

1	1	0	0
1	1	1	1
0	0	0	0
0	0	1	0

3	2	1	0

we want

→

2	0	4	1

2	4	0	1

4	2	0	1

③

4	2	1	0

0 0 0 0
1 0 0
1 1 1 0 → 0
0 0 1 0

Final

n=4

→

1	2	0	3

→ 0 1 2 3

3	2	1	0

0 1 2 3

n-i-1

4-0-1 = 3
4-1-1 = 2
4-2-1 = 1
4-4-0 = 0

n-i-1 = 3 ①

1 2 3 0

1 3 2 0 ③

100-1

3 1 2 0

i=0
i=1

②

3 2 1 0

1

4 =

1	2	1	0

-1

n=4

n-i-1

-1

0 0 0 1 → 0
1 1 0 0 → 2
1 0 1 0 → 1
0 0 1 0 → 1

Dry Run :-

[[0 0 1 0] →
[1 1 0 0] →
[1 0 0 0] →
[1 1 1 0]] →

Req = [3, 2, 1, 0]

(n-i-1)

Step 1 :- Trailing zero Count :-

Row	Binary Row	Trailing Zero
0	[0 0 1 0]	1
1	[1 1 0 0]	2
2	[1 0 0 0]	3
3	[1 1 1 0]	1

arr = [1 | 2 | 3 | 1]
0 1 2 3

Step	i	Required (n-i-1)	arr Before (0 1 2 3)	Found Idx	Add swap	arr After	ans
1	0	3	[1, 2, 3, 1]	2 (val = 3)	2	[3, 1 2 1]	2
2	1	2	[3, 1, 2, 1]	2 (val = 2)	1	[3, 2, 1, 1]	3
3	2	1	3 2 1 1	//	//	3 2 1 1	3
4	3	0	3 2 1 1	//	//	3 2 1 1	3

→ 1 2 3 1 | 3 1 2 1
1 3 2 1 | 3 2 1 1
3 1 2 1

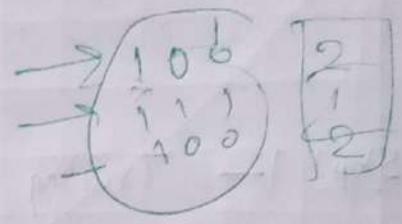
=

Return Ans = 3

Algo :- Flow :-

Step 1 :- Count trailing zero

1. n = grid.length.
2. arr[n]
3. Har row ke liye
   - Last column se left jao
   - Jab tak 0 mile count++
   - Jere 1 mile X
   - Us row ka trailing zero arr[i] store kro.

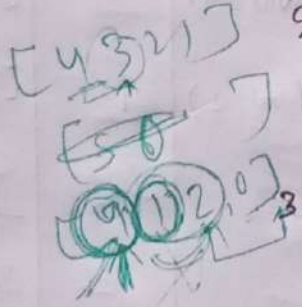


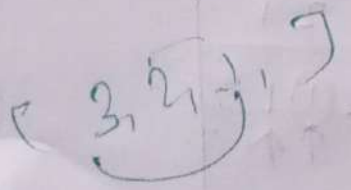
arr[i] = row i me kitne trailing 0 hai.

Step 2 :- Har position ko Greedy fix karo

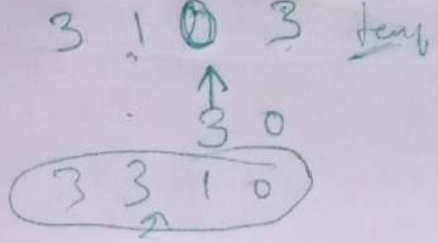
i ⟶ 0 .... n-1

1. Required zero calculate karo :-
   req = n-i-1
2. Agar arr[i] ≥ req
   OK
   Continue.
3. Nahi to neche search karo (j = i+1 → n-1)
   ⇒ Pehli row dhundo jisme arr[j] ≥ req.
4. Agar aisi row mil jaye :-
   Swap count badhao :-
   ans + (j-i)
   Us row ko bubble shift karo
   arrayre (arr, i, j)
5. Agar aisi row nahi mili
   → Return -1





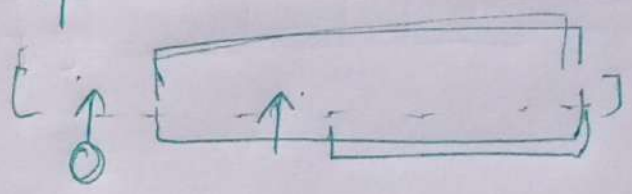
Step 3 :- return ans.



T.C = $O(n^2)$

S.C = $O(n)$

2D → 1D → arr

n×n

[1 0 2 3]